

Billing & Folio Module v1

Status

LOCKED – Billing v1

This document describes the **Billing & Folio v1** behavior, operational runbook, and test cases for the Hospitality SaaS backend.

Scope (v1)

Included: - Manual folio creation (ADMIN) - Folio retrieval by reservation - Line items (service charges, discounts, adjustments) - Role-based restrictions - Folio close lifecycle - Rate plan based pricing (INR)

Explicitly **NOT included** in v1: - Auto room-night posting - Invoicing / PDF - Taxes / GST - Payments

Domain Concepts

Folio

A folio represents the financial ledger for a single reservation.

Constraints: - 1 folio per reservation - Immutable after CLOSED

Line Items

Atomic financial entries inside a folio.

Supported types: - ROOM_CHARGE (reserved) - SERVICE_CHARGE - MINIBAR - ADJUSTMENT (ADMIN only) - DISCOUNT (ADMIN only)

Roles & Permissions

Action	ADMIN	STAFF
Create folio	✓	✗
View folio	✓	✓
Add service charge	✓	✓
Add discount	✓	✗
Close folio	✓	✗

API Runbook

1. Create Rate Plan (Prerequisite)

```
POST /rate-plans?tenantCode=SUNU_HOTEL
```

2. Create Reservation

```
POST /reservations
```

3. Confirm → Check-in → Check-out

```
PATCH /reservations/:id/status
```

Lifecycle:

```
PENDING → CONFIRMED → CHECKED_IN → CHECKED_OUT
```

4. Create Folio (ADMIN)

```
POST /folios
{
  "reservationId": "<uuid>"
}
```

5. Add Line Item

```
POST /folios/:folioId/line-items
```

Example (STAFF allowed):

```
{
  "type": "SERVICE_CHARGE",
  "description": "Airport Pickup",
  "quantity": 1,
  "unitPrice": 75,
  "currency": "INR"
}
```

6. Close Folio (ADMIN)

```
PATCH /folios/:folioId/close
```

Validation Rules

- Cannot add line items to CLOSED folio
 - Discounts & adjustments require ADMIN
 - Folio is unique per reservation
 - Currency fixed to INR (v1)
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Test Cases

TC-01 Create Folio

- Given reservation exists
- When ADMIN creates folio
- Then folio status = OPEN

TC-02 Duplicate Folio

- Given folio exists
- When ADMIN creates again
- Then 409 Conflict

TC-03 Staff Add Service Charge

- Given folio OPEN
- When STAFF posts SERVICE_CHARGE
- Then line item created

TC-04 Staff Add Discount

- Given folio OPEN
- When STAFF posts DISCOUNT
- Then 403 Forbidden

TC-05 Admin Add Discount

- Given folio OPEN
- When ADMIN posts DISCOUNT
- Then line item created

TC-06 Close Folio

- Given folio OPEN
- When ADMIN closes folio
- Then status = CLOSED

TC-07 Mutate Closed Folio

- Given folio CLOSED
 - When adding line item
 - Then 400 Bad Request
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Known Limitations (v1)

- Room charges are manual / deferred
 - No tax or GST computation
 - No invoice generation
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Next Planned (v2)

- Auto room-night charges on CHECKED_OUT
 - Invoice entity
 - Tax / GST layers
 - Payment settlement
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Final Note

Billing v1 is **stable, isolated, and production-safe**. No cross-module side effects exist.